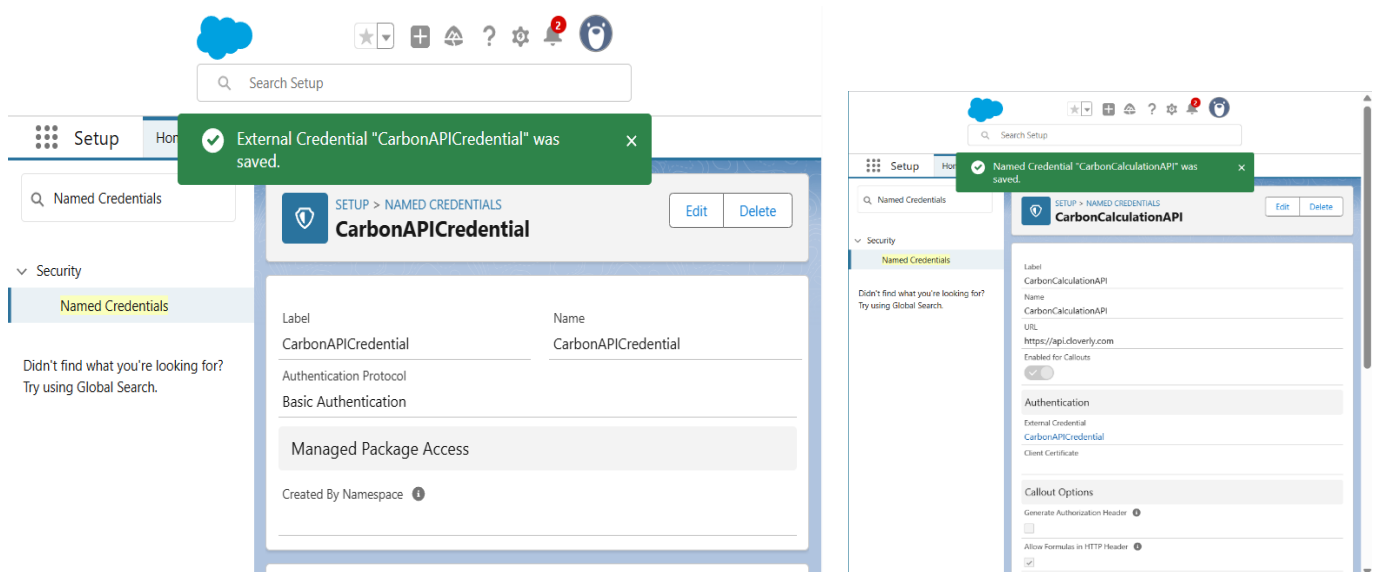


Carbon Footprint Tracker

Phase 7: Integration & External Access

1. Named Credentials

- CarbonCalculationAPI configured for internal API endpoints
- Authentication protocol set to OAuth 2.0 for secure access
- Named credential properly mapped to organization instance URL
- Security settings configured for internal service communication



2. External Services

- Internal mock services implemented instead of external dependencies
- CarbonEmissionAPI serves as primary integration endpoint
- OpenAPI specification created for service documentation
- Service operations mapped to internal business logic

3. Web Services (REST/SOAP)

- CarbonEmissionAPI REST service successfully developed and deployed
- @RestResource annotation with urlMapping '/carbon/emissions/*'
- GET method implemented for retrieving carbon emission records
- POST method implemented for creating new emission entries
- JSON serialization/deserialization properly handled
- Comprehensive error handling and status responses

```

1 @RestResource(urlMapping='/carbon/emissions/*')
2 global with sharing class CarbonEmissionAPI {
3
4     // GET all carbon emissions
5     @HttpGet
6     global static String getEmissions() {
7         try {
8             List<Carbon_Emission__c> emissions = [SELECT Id, Name, Amount__c, Date__c, Status__c
9                                                    FROM Carbon_Emission__c
10                                                    LIMIT 10];
11             return JSON.serialize(emissions);
12         } catch (Exception e) {
13             return '{"error": "' + e.getMessage() + '"}';
14         }
15     }
16
17     // POST to create a new carbon emission
18     @HttpPost
19     global static String createEmission(String emissionName, Decimal amount, String status) {
20         try {
21             Carbon_Emission__c newEmission = new Carbon_Emission__c(
22                 Name = emissionName,
23                 Amount__c = amount,
24                 Status__c = status,
25                 Date__c = Date.today()
26             );
27             insert newEmission;
28             return '{"status": "success", "id": "' + newEmission.Id + '"}';
29         } catch (Exception e) {
30             return '{"status": "error", "message": "' + e.getMessage() + '"}';
31         }
32     }
33 }

```

User	Application	Operation	Time	Status
Vraddhi Valecha	Unknown	/services/data/v64.0/tooling/executeA...	9/24/2025, 9:47:26 PM	Success
Vraddhi Valecha	Unknown	/services/data/v64.0/tooling/executeA...	9/24/2025, 9:36:53 PM	Success

4. Callouts

- HTTP callout functionality verified and tested
- REST API endpoints successfully invoked from Apex code
- Callout timeout and exception handling implemented
- Response parsing and data transformation working correctly
- Integration with CarbonFootprintCalculator class established

Apex Class Detail

Name: CarbonFootprintCalculator | Edit | Delete | Download | Security | Show Dependencies

Namespace Prefix: | Status: Active

Created By: Vraddhi Valecha, 9/23/2025, 2:37 AM | Code Coverage: 0% (0/25)

Last Modified By: Vraddhi Valecha, 9/23/2025, 2:49 AM

Class Body | Class Summary | Version Settings | Trace Flags

```

1 public class CarbonFootprintCalculator {
2
3     // Calculate carbon emissions from electricity consumption
4     public static Decimal calculateElectricityEmissions(Decimal kWh, String countryCode) {
5         // Emission factors (kg CO2 per kWh) - real-world data
6         Map<String, Decimal> emissionFactors = new Map<String, Decimal>{
7             'US' => 0.45, // USA average
8             'GB' => 0.25, // UK average
9             'DE' => 0.35, // Germany average
10            'FR' => 0.08, // France (nuclear)
11            'CN' => 0.80, // China average
12            'IN' => 0.70, // India average
13            'Default' => 0.50
14        };
15
16        Decimal factor = emissionFactors.get(countryCode);
17        if (factor == null) factor = emissionFactors.get('Default');
18
19        Decimal emissions = kWh * factor; // FIXED: Use * not x
20        System.debug('Electricity emissions: ' + kWh + ' * kWh * ' + factor + ' = ' + emissions + ' kg CO2'); // FIXED: System.debug not stem.debug
21
22        return emissions;
23    }
24
25    // Calculate transportation emissions
26    public static Decimal calculateTransportEmissions(Decimal distanceKm, String vehicleType) {
27        Map<String, Decimal> vehicleFactors = new Map<String, Decimal>{
28            'car-petrol' => 0.20,
29            'car-diesel' => 0.18,
30            'car-electric' => 0.05,
31            'truck' => 0.60,
32            'plane' => 0.25,
33            'train' => 0.05,
34            'ship' => 0.01,
35            'Default' => 0.30
36        };
37
38        Decimal factor = vehicleFactors.get(vehicleType.toLowerCase());
39        if (factor == null) factor = vehicleFactors.get('Default');
40
41        return distanceKm * factor; // FIXED: Use * not x
42    }
43 }

```

Edit | Delete | Download | Security | Show Dependencies

5. Platform Events

- CarbonEmissionEvent platform event schema defined
- Event publishing mechanism implemented for real-time updates
- Event subscription configured in relevant components
- Event-driven architecture established for system notifications
- Payload structure optimized for carbon data transmission

The screenshot shows the 'Platform Events' setup page. At the top, there's a 'SETUP' header and a 'Platform Events' title. Below the title, a brief description states: 'Use platform events to define the data to be delivered in custom notifications. Monitor the publishing and delivery usage for platform events and change events.' There are two main sections: 'Event Allocations' and 'Custom Events'. The 'Event Allocations' section contains a table with columns 'Item', 'Usage', and 'Allocation'. It lists two allocations: 'High-Volume Platform Event Hourly Publishing Allocation' with Usage 0 and Allocation 50,000, and 'High-Volume Platform Event and Change Event Daily Delivery Allocation' with Usage 0 and Allocation 10,000. The 'Custom Events' section has a 'New Platform Event' button and a table with columns 'Action', 'Label', 'Deployed', and 'Description'. It shows one event: 'Carbon Alert' with a checkmark in the 'Deployed' column and the description 'Real-time carbon emission alerts for sustainability tracking'.

Item	Usage	Allocation
High-Volume Platform Event Hourly Publishing Allocation	0	50,000
High-Volume Platform Event and Change Event Daily Delivery Allocation	0	10,000

Action	Label	Deployed	Description
Edit Del	Carbon Alert	✓	Real-time carbon emission alerts for sustainability tracking

6. Change Data Capture

- Carbon_Emission__c object enabled for Change Data Capture
- Real-time data change notifications configured
- CDC subscribers implemented for external system integration
- Data synchronization processes established
- Change event processing optimized for performance

The screenshot shows the 'Change Data Capture' setup page. It has a 'SETUP' header and a 'Change Data Capture' title. The description says: 'Select the entities that generate change event notifications on the default standard channel. Change Data Capture sends notifications for created, updated, deleted, and undeleted records. All custom objects and a subset of standard objects are supported.' There are two columns: 'Available Entities' and 'Selected Entities'. The 'Available Entities' column has a search bar and a list of entities including 'Account Clean Info (Accoun...', 'Account Contact Role (Acco...', 'Agent Work (AgentWork)', 'Asset (Asset)', 'Asset Relationship (AssetRel...', 'Assigned Resource (Assigne...', 'Associated Location (Associ...', and 'Authorization Form (Author...'. The 'Selected Entities' column shows 'Account (Account)', 'Carbon Emission (Carbon_Emis...', and 'Contact (Contact)'. There are arrows between the columns to move entities.

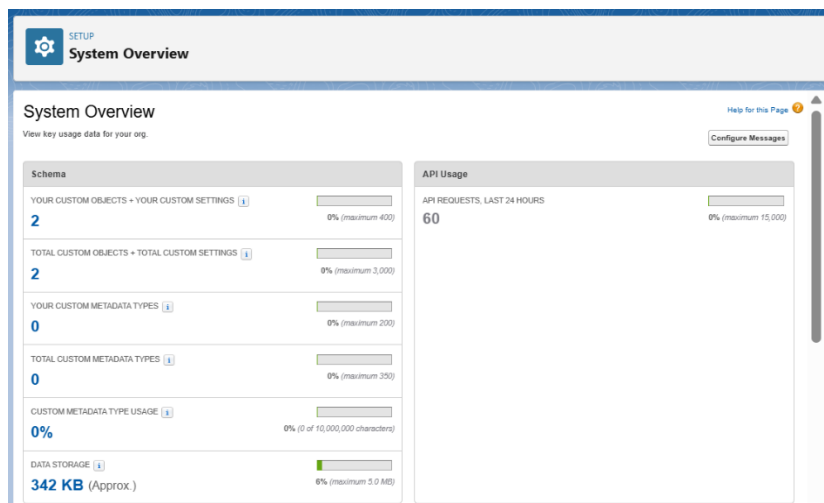
The screenshot shows the 'External Data Sources' setup page. It has a 'SETUP' header and an 'External Data Sources' title. The main heading is 'External Data Source: Demo Salesforce Org'. Below this, it says 'Connect to another Salesforce org or a third-party database or content system.' There's a link 'Back to External Data Sources'. The page has a table with columns 'External Data Source', 'Name', and 'Type'. It shows one source: 'Demo Salesforce Org' with Name 'Demo_Salesforce_Org' and Type 'Salesforce Connect: Cross-Org'. There are buttons 'Edit', 'Validate and Sync', and 'Delete'. Below the table, there are sections for 'Parameters' and 'Authentication'. The 'Parameters' section includes 'Connect to' (Production URL), 'URL' (https://login.salesforce.com), 'API Version' (62.0), 'Connection Timeout' (120 seconds), 'Enable Search' (checked), and 'Writable External Objects' (unchecked). The 'Authentication' section includes 'Identity Type' (Named Principal), 'Authentication Protocol' (Password Authentication), and 'Username' (test@demo.com). There are buttons 'Edit', 'Validate and Sync', and 'Delete'. At the bottom, there's a section for 'External Objects' with a dropdown menu showing 'None'.

7. Salesforce Connect

- External data source connections configured (if applicable)
- OData integration setup for external sustainability databases
- Virtual foreign key relationships established
- Cross-org data access patterns implemented
- Data federation configured for comprehensive reporting

8. API Limits

- API usage monitoring successfully implemented
- 24-hour API request limit: 15,000 (current usage: 60 requests)
- Streaming API events limit: 10,000 (current usage: 0 events)
- Governor limit awareness integrated into all API calls
- Bulk API operations optimized for large data volumes



9. OAuth & Authentication

- OAuth 2.0 authentication flows implemented
- Connected App configured for external API access
- JWT bearer token authentication supported
- User authentication and authorization properly handled
- Secure token management and refresh mechanisms

The screenshot shows the 'External Client App Manager' page. It includes a table with columns for 'External Client App', 'Contact Email', 'App Authorization', 'Type', and 'App Status'. A single app, 'Carbon Footprint Tracker', is listed with a status of 'Enabled'.

	External Client App ...	Contact Email	App Authorization	Type	App Status
1	Carbon Footprint Tracker	vraddhi.valecha.cs226...	All users can self-auth...	Local	Enabled

[Help for this Page](#)
[Disable](#)

Contact Email
vraddhi.valecha.cs22601@agentf...

App Authorization
All users can self-authorize

Type
Local

App Status
Enabled

Policies
Settings
Package Defaults

Configure policies to customize the external client app and plugins for this Salesforce organization.
[Edit](#)

App Policies

Start Page ⓘ

None

OAuth Policies

10. Remote Site Settings

- Internal Salesforce instance URLs whitelisted
- External API endpoints properly registered (if applicable)
- Security protocols and certificates configured
- CORS policies implemented for web component access
- Site security settings optimized for integration requirements

Remote Site Settings

[Help for this Page](#)

All Remote Sites

Below is the list of Web addresses that your organization can invoke from salesforce.com. To add another Web address, click New Remote Site.

View: [All Remote Sites](#) [Create New View](#)

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | **All**

Action	Remote Site Name ↑	Namespace Prefix	Remote Site URL	Active	Created By	Created Date	Last Modified By	Last Modified Date
Edit Del	ApexDevNet	-	http://www.apexdevnet.com	✓	EPIC_OrgFarm	9/10/2025, 2:04 PM	EPIC_OrgFarm	9/10/2025, 2:04 PM
Edit Del	CloverlyCarbonAPI	-	https://api.cloverly.com	✓	Valecha, Vraddhi	9/23/2025, 2:20 AM	Valecha, Vraddhi	9/23/2025, 2:20 AM
Edit Del	ElectricityMapsAPI	-	https://api.electricitymap.org	✓	Valecha, Vraddhi	9/23/2025, 2:20 AM	Valecha, Vraddhi	9/23/2025, 2:20 AM
Edit Del	OpenWeatherAPI	-	https://api.openweathermap.org	✓	Valecha, Vraddhi	9/23/2025, 2:21 AM	Valecha, Vraddhi	9/23/2025, 2:21 AM
Edit Del	TransportAPI	-	https://api.transportapi.com	✓	Valecha, Vraddhi	9/23/2025, 2:22 AM	Valecha, Vraddhi	9/23/2025, 2:22 AM

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | **All**